

NEAR-SCALPER-BOT

Complete Strategy Dossier — Both Engines, Honest Assessment

Prepared for the owner's independent research — refine it, or replace it.

Date: 12 September 2026 · Live since: 11 September 2026 (07:20 UTC first trade) · Repo: CozanetHQ/near-scalper-bot · Account: paper, \$10 session trial (reset 11 Sep 18:05 UTC)

1. Executive Summary — the verdict up front

The system is a **1-minute EMA pullback scalper** (Engine 1) with an **advisory structure-scoring engine** (Engine 2) that is recorded but never acts. Engine 1 trades 5 pairs (BTC, ETH, SOL, XRP, NEAR) on exchange paper prices, up to 8 concurrent slots, 10x leverage, with layered damage-control exits.

The honest verdict: the chassis is excellent, the edge is not yet proven. Across 196 lifetime trades the strategy has a slightly negative expectancy (about $-\$0.006$ per trade). It wins 90%+ of trades for pennies (average TP pays $+\$0.023$) and loses 16.8% of trades at $-\$0.079$ average ($-\$0.44$ each in the current \$10 era). The risk:reward is roughly **1 : 0.29** — the survival margin is thin, and profitability currently depends on the loss-control machinery, not on the signal.

Where the edge candidate actually lives: the strategy's results are **strongly regime-dependent**. NEAR was net-positive with zero kills in the 08:00–12:00 UTC trend window and lost heavily in midday chop. The signal is not garbage — it is a trend-pullback logic being applied in all conditions, including ones it was never meant to trade. The regime-gating experiments now in flight are the decision point between refinement and replacement.

2. System architecture in one page

A GitHub Actions cron tick (every ~4 minutes) runs tick.py: fetch 1m + 15m candles for each of the 5 pairs, manage open positions (exits), then evaluate new entries per pair. All locked parameters live in param_registry.json (v4 discipline: every constant carries a rationale and lock date; a lineage mismatch between code and registry blocks new entries automatically). Per-pair overrides live in the registry's per_pair section and resolve as: lab env override > registry per_pair > global default.

Data layers: state.json (positions ride inside a whitelist-safe field), trades.jsonl (the permanent ledger — now recording regime_at_entry and atr_frac_at_entry on every new trade), and a public GitHub Pages dashboard.

Governance layer (constitution): AI advises, human authorizes, then AI executes. No rule is locked without the owner's sign-off; no rule is global across pairs — everything is tested and locked per pair.

3. Engine 1 — the main trading engine (the scalper)

3.1 Signal (what makes it want to enter)

On 1-minute candles: **EMA9 vs EMA21 momentum** defines direction. Entry wants a **pullback candle against that momentum** — a bearish candle inside an EMA-bull regime with price still above EMA21 (buy the dip), or the mirror for shorts (sell the rip). It is a classic trend-continuation micro-scalp: it never counter-trends against a runaway, and it never enters on a spike.

Gate conditions that must all pass at the moment of entry:

- **ATR floor:** $1m\ ATR \geq 0.08\% \text{ of price (day)} / 0.20\%$ (21:00–07:00 UTC — thin-session chop must pay more)
- **Regime filter:** if the last 6 closed 15m candles are ALL one color ("runaway"), counter-trend entries are blocked. If the forming 1m candle range exceeds $3 \times ATR$ ("spike" — manipulation/liquidation cascade risk), all entries blocked this tick

- **EV-after-costs gate:** p_win is blended 0.85 prior (weight 20 pseudo-trades) with observed wins/losses; $EV = p \cdot (tp_frac - 2 \cdot fees) - (1-p) \cdot 2\%$ assumed loss – slippage; entries require $EV \geq +0.05\%$ per unit, else WAIT
- **Cooldown:** 90 seconds minimum between entries — one signal cluster cannot fill all slots

3.2 Position sizing

Account balance $\times$ 85% margin budget $\div$ 8 slots = per-slot margin; notional = margin $\times$ 10x leverage. A single trade can never hog the budget (the 2026-09-06 wedge lesson). Target win per trade: \$0.05, i.e. $\sim 1.67\%$ of balance (WIN_TARGET_PCT 0.0167). Below the 25% drawdown wall, sizing drops to 1/4 (recovery mode, floor \$1) — the account never fully pauses, because a paused account can never recover (Phase 4 finding).

3.3 Exits — the damage-control stack (in evaluation order per tick)

1. **TP:** take-profit at $1.6 \times 1m$ ATR, floored at 0.6% of price (below that, fees eat the scalp). Optional swing-TP mode (currently OFF) front-runs 15m fractal swing levels instead.
2. **BE_STOP (dynamic stop):** once MFE reaches 50% of the TP distance (the empirically validated cohort — 91% of those went on to hit TP in the seed), the trade exits at breakeven+costs if price retraces. This is why 51% of all trades are scratches that still cover fees.
3. **MAE ceiling (per-pair "wall"):** if adverse excursion exceeds the pair's locked wall, exit immediately — beyond that point recovery odds are gone. Global 4.0%; NEAR tightened to 2.5% on 12 Sep (owner-directed sim).
4. **HARD_SL 6%:** the a-priori backstop (locked before any results; sits inside the $\sim 9.5\%$ liquidation point at 10x).
5. **MAX_POS_AGE 48h:** hard market-close of any position older than the pair's cap (per-pair overrides proposed: BTC 8h, ETH 8h, SOL 6h — pending owner authorization).
6. **TP-aging:** a position stuck for hours relaxes its TP target toward entry (50% at 6h, 75% at 24h, 90% at 72h) — it never crosses below entry-plus-costs; it just stops demanding full profit to free the slot.

3.4 Account-level risk machinery

- **12% daily loss limit:** after losing 12% from the UTC day-start balance, no new entries for the rest of the day (hard-wired, registry-locked)
- **25% drawdown wall:** 1/4 sizing while below it (recovery mode)
- **Per-pair slot caps:** global budget is 8 slots; NEAR capped at 3 (locked 12 Sep — its 4+ deep-stack bucket carried 11 of its 14 lifetime kills)
- **Slot recycle:** wedged trades don't stop the book — TP-aging + age caps keep slots circulating
- **No averaging down, ever** (owner principle 1). Recovery logic manages existing positions only.

3.5 Cost model

Fees 0.06% per side (taker both ways) + configurable slippage stress per side (currently 0 in live, used in lab). Every BE_STOP exit is engineered to cover round-trip costs — scratches are engineered to be free, not winners.

3.6 What Engine 1 does NOT do (honest gaps)

- No order-flow, volume, funding, or OI data — pure price action on 1m/15m
- No intrabar granularity: decisions happen once per tick on candle closes; fills are assumed at the current price
- The regime classifier only labels runaway/spike — **it does not yet gate on "chop"**, which the data says is the expensive condition
- No take-profit asymmetry: winners are capped at $\sim 1.6 \times ATR$ while losers can run to the wall — the R:R inversion documented in section 5

4. Engine 2 — the advisory second engine

A pure-function structure scorer, computed on 15m fractal swings at the moment Engine 1 fires a signal. **It is advisory only: it records its scores into the trade ledger but never gates, blocks, or sizes anything, and**

there is no feedback path into execution. Three components:

- **Structure quality (weight 0.4):** fractal swing analysis on the last 40 15m candles. Scores a market-structure read in the trade's direction: Break of Structure +0.15, Staircase (two higher same-direction swings) +0.40, Change of Character (close beyond last opposing swing) +0.70, Market-Structure-Shift (beyond two opposing swings) +1.00, none 0.00
- **Move potential (weight 0.4):** distance to the nearest opposing fractal wall vs the TP distance. Wall $\geq 1.3 \times$ TP away: 0.90 (room to run). Wall $< 0.7 \times$ TP: 0.15 (target sits beyond the wall — likely rejection). A high spike-risk bucket caps this at 0.20
- **Spike risk (weight 0.2, inverted):** forming 1m range $\div$ ATR $\div 3$ — the manipulation/liquidation-cascade detector

Overall score = $0.4 \cdot (2 \cdot \text{structure} - 1) + 0.4 \cdot (2 \cdot \text{move} - 1) + 0.2 \cdot (1 - 2 \cdot \text{spike})$, clamped to $[-1, +1]$. Every trade carries structure_quality, move_potential, wall_ratio, spike_bucket, and overall_score in the ledger.

Honest assessment of Engine 2: it has been recording scores since 11 Sep but nothing has yet been done with them — no validation that its scores correlate with outcomes, no thresholds locked. It is a hypothesis machine, not a working filter. The validation study (score vs. outcome buckets) is pending; the owner can decide whether it earns a gating role or gets replaced. Its one proven component is dist_to_wall (validated in lab).

5. The bet structure — the honest math

Every trade this system has ever taken (196 trades, lifetime net -0.81 on a paper account that has run at \$3 and \$10):

Exit type	Trades	% of all	Avg P&L	Total P&L
TP	62	31.6%	+0.0232	+1.44
BE_STOP	101	51.5%	+0.0035	+0.35
MAE_KILL	33	16.8%	-0.0789	-2.60

The bet, distilled: **TP pays ~+\$0.023 average (31.6% of trades); the MAE wall costs ~-\$0.079 average lifetime (-\$0.44 each in the current \$10 era, 16.8% of trades); everything else is engineered to be ~zero.**

Risk:reward $\approx 1 : 0.29$. For that payoff the kill rate must stay under **22.7%** just to break even; actual is 16.8%. That is the whole margin: the strategy survives on its damage control, with per-trade EV of roughly **-\$0.006** before the 12 Sep containment locks (2.5% NEAR wall halves each NEAR kill; the 3-slot cap removes the bucket that produced 11 of NEAR's 14 kills).

Reading this honestly: a 90% win rate is worthless if the payoff is 1:0.29 — it is cosmetic. What would flip the EV positive: (a) a higher TP/kill ratio (bigger targets, or cheaper kills), or (b) skipping the conditions that produce the kills (regime gating). Both are testable with the data now being recorded.

6. What it has actually done — track record & timeline

Two data eras — **quality disclosure:** before 11 Sep 15:06 UTC a telemetry bug ran every pair's MAE calculation on NEAR candles, so cross-pair kill attributions in Era 1 are contaminated (NEAR's own numbers were correct; other pairs' "kills" were partly bogus). All decisions since are made on clean-since-fix data.

Era timeline:

- **10 Sep:** v5 upgrade — EV-after-costs gate, regime classifier, account reset to \$3, all params locked a-priori in the registry (v4 discipline)
- **11 Sep morning:** first NEAR era — 100% WR over the first 20 trades (trend window)
- **11 Sep 15:06:** the cross-pair candle bug found and fixed; clean data begins
- **11 Sep afternoon:** 30-day 5-pair validation; per-pair registry architecture built (age-cap grid: no single cap fits all pairs — universal rules withdrawn)

- **11 Sep 18:05:** account hard-reset to \$10 for the session trial; dashboard rebuilt institutional-grade; ledger/balance chain-guard fixed (mid-flight trades excluded from session accounting)
- **11 Sep evening:** two NEAR long MAE_KILLS at the 4% wall (−\$0.88 together) eat the session's wins
- **12 Sep 01:00–05:00:** containment locks — NEAR MAE wall 2.5% (owner-directed sim), NEAR slot cap 3 (concurrency study), regime_at_entry + atr_frac_at_entry telemetry wired into every new trade

Per-pair record (lifetime, with era caveat):

Pair	Trades	Net P&L	WR	TPs	MAE_KILLS	Era-1 caveat
NEAR	138	-0.9730	90%	59	14	own data clean
ETH	12	-0.0203	50%	0	8	kills partly bogus
BTC	5	+0.0110	60%	0	2	kills partly bogus
XRP	26	+0.0726	81%	2	5	kills partly bogus
SOL	15	+0.0971	73%	1	4	kills partly bogus

Clean era only (since the fix, 11 Sep 15:06 UTC):

Pair	Trades	Net P&L	WR	TPs	MAE_KILLS
SOL	10	+0.1173	100%	1	0
XRP	8	+0.0586	100%	1	0
ETH	4	+0.0299	100%	0	0
BTC	3	+0.0203	100%	0	0
NEAR	50	-1.1232	84%	19	8

The NEAR deep-dive (138 trades, the pair with the real evidence):

Win rate degraded chronologically: 100% → 90% → 95% → 90% → 85% → 90% → 78% per 20-trade window, kills accelerating 0→2→1→2→3→2→4. By entry hour (UTC): 08:00–11:59 was +\$0.37 across 36 trades with **zero** kills; 12:00–16:00 bled −\$0.60 with 11 kills. By concurrency at entry: the 3-slot bucket was its best (+\$0.25, 100% WR, 0 kills) while the 4+ deep-stack bucket (95 trades) carried 11 of 14 lifetime kills (−\$0.52). Same strategy, opposite outcomes by regime and stack depth — the strongest evidence the signal is regime-dependent rather than worthless.

Current session (\$10 trial, started 11 Sep 18:05 UTC): \$10.00 → \$9.45 at the time of writing (−5.5%), 30 trades, 2 kills — both before the containment locks. Since the locks went live: zero kills, small positive drift, locks not yet stress-tested. The 12% daily limit is armed every UTC day.

7. Strengths and weaknesses — no varnish

Genuine strengths:

- Damage control is best-in-class for this scale: BE_STOP rescues half of all trades at breakeven+costs; walls, hard SL, age caps, daily limit, per-pair locks — every one empirically derived and registry-locked with a rationale and date
- Parameter discipline: a-priori locking, lineage checks that block entries on mismatch, per-pair resolution, no tuning on results (the wall and cap were chosen from sims over history, then locked)
- Ledger quality: per-trade MFE/MAE, duration, entry context, and now regime/ATR at entry — every decision is auditable, every rule has its evidence trail
- Zero catastrophic failures: no account has ever blown through the machinery; worst single-session loss was contained to single-digit percents

Genuine weaknesses:

- **The R:R inversion is structural:** winners are capped at $\sim 1.6 \times 1m$ ATR ($\approx 0.2\text{--}0.3\%$ moves) while the kill path runs to 2.5–6%. No exit engineering can fix a payoff that asymmetric — the TP side or the kill frequency must move
- **Chop is not gated:** the classifier blocks runaway and spike, but "chop" entries pass and die. The midday data proves it. This is the single biggest known leak
- **The advisory engine is unused:** scores accumulate, nothing consumes them — no proven correlation with outcomes yet
- **Tick-granular execution:** ~ 4 -minute decision cadence on candle closes with assumed fills — fast moves between ticks are modeled only through stress slippage
- **No volume/order-flow confirmation:** pure price action; pullback quality is judged by candle color and EMA position only
- **Sample sizes are still thin:** 196 trades total, ~ 22 hours of live life; every conclusion above is provisional and the study pipeline exists because of that

8. Open experiments & decision criteria (what is being waited on)

1. **Chop-regime entry gate** — needs 50–100 regime-tagged trades (accumulating now, telemetry live since 12 Sep). Decision rule: if gating chop entries moves the kill rate materially toward single digits without giving up the TP count, lock it per pair. If the kill rate barely moves, the signal is the problem — replace the strategy, don't gate it.
2. **TP-sizing / payoff asymmetry** — the 1:0.29 R:R is the root defect. Candidates: swing-level TP targets (the engine already has the mode, OFF), partial profit capture with a runner, or wider SCALP_TP_ATR. Needs its own backtest on the two-window lab data before any lock.
3. **Engine 2 validation** — bucket every ledger trade by advisory score; if overall_score discriminates winners from kills, it earns a gating role at a locked threshold; if not, retire it or replace its features.
4. **Age-cap per-pair locks** — already validated two-window (BTC 8h, ETH 8h, SOL 6h, XRP/NEAR keep 48h) and sitting in the registry awaiting owner authorization.

9. For your own research — where to look

- **The raw evidence:** data/trades.jsonl (every trade with full context), param_registry.json (every rule with its rationale and lock date), research/phase4_validation_2026-09-11.md (the sims behind the wall/cap locks), tick.py + engine/ (all logic, commented with the history of why)
- **The key question to answer first:** is the 1m EMA-pullback signal edge-positive in trending conditions alone? If yes → the refinement path is regime gating + payoff asymmetry. If no → the signal itself is the problem and the machinery deserves a better brain
- **Benchmarks to compare against:** momentum-pullback continuation strategies with asymmetric exits; the payoff-asymmetry literature ("cutting winners" studies); 1m-scalp fee-drag analyses
- **Cost reality check:** fees at 0.06%/side mean $\sim 0.12\%$ per round trip — over half of the average TP win. Any refined strategy must treat fee drag as a first-class constraint
- **What I would test first if I were you:** re-run the lifetime ledger with a simulated chop gate and a swing-TP (both are single registry/env changes in the lab, no code changes needed)

— Prepared by Solene · every number in this document is reproducible from the repo's own ledger and code.